Lecture 2

Classification

The problem of prediction a discrete random variable $\mathbf{y}$ from another random variable $\mathbf{x}$ is Called classification. Consider *iid* data $(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_n, \mathbf{y}_n)$ where

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ \vdots \\ x_{id} \end{bmatrix} \in \mathcal{X} \subset \mathbb{R}^d$$

is a $d - dimensional$ vector and $\mathbf{y}_i$ takes values in some finite set $\mathcal{Y}$. A classification rule is a function $h : \mathcal{X} \to \mathcal{Y}$. When we observe a new $\mathbf{x}$ we predict $\mathbf{y}$ to be $h(\mathbf{x})$.

Error rate

Definition: The true error rate of a classifier h is

$$L(h) = Pr(h(\mathbf{x}) \neq \mathbf{y})$$

and the empirical error rate or training error rate is:

$$\hat{L}_n = \frac{1}{n} \sum_{i=1}^{n} I(h(\mathbf{x}_i) \neq \mathbf{y}_i)$$

Bayes Classifier

Consider the special case where $\mathcal{Y} = \{0, 1\}$ then

$$\begin{aligned} r(x_0) &= P(\mathbf{y} = 1 | \mathbf{x} = x_0) \\ &= \frac{P(\mathbf{x} = x_0 | \mathbf{y} = 1) P(\mathbf{y} = 1)}{P(\mathbf{x} = x_0)} \\ &= \frac{P(\mathbf{x} = x_0 | \mathbf{y} = 1) P(\mathbf{y} = 1)}{P(\mathbf{x} = x_0 | \mathbf{y} = 1) P(\mathbf{y} = 1) + P(\mathbf{x} = x_0 | \mathbf{y} = 0) P(\mathbf{y} = 0)} \end{aligned}$$

Bayes Classifier

Definition: The Bayes classification rule h^* is

$$h(x_0) = \begin{cases} 1 & \textit{if } \hat{r}(x_0) > 1/2 \\ 0 & \textit{otherwise} \end{cases}$$

Decision Boundary

The set $D(h) = \{\mathbf{x} : P(\mathbf{y}=1|\mathbf{x}=x_0) = P(\mathbf{y}=0|\mathbf{x}=x_0)\}$ is called the decision boundary.

$$h(x_0) = \begin{cases} 1 & if \;\; P(\mathbf{y}=1|\mathbf{x}=x_0) > P(\mathbf{y}=0|\mathbf{x}=x_0) \\ 0 & otherwise \end{cases}$$

Bayes Classifier

Theorem: The Bayes rule is optimal, that is if h is any other classification rule then $L(h^*) \leq L(h)$.

Bayes Classifier

Why do we need any other method?

Bayes Classifier

The Bayes rule depends on unknown quantities, so we need to use the data to find some approximation to the Bayes rule.

From Bayes Rule to Learning

The optimal binary classifier is

$$h^*(x) = 1\{\,\eta(x) > 1/2\,\}, \quad \eta(x) = P(Y = 1 \mid X = x)$$

- The posterior probability η(x) is unknown.
- We observe only a finite training sample, so we must learn an approximation from data.
- **The key distinction is what statistical object the algorithm learns.**

Three Routes to a Classifier

1. **Learn class-conditional densities and priors**

   Estimate p(x | y) and P(y); then use Bayes' rule.

   $p(x \mid y),\ P(y) \ \rightarrow\ P(y \mid x) \ \rightarrow\ h(x)$

   Often called the generative route.

2. **Learn posterior probabilities directly**

   Estimate P(y | x) without modeling the distribution of X.

   $P(y \mid x) \ \rightarrow\ h(x) = \arg\max_{\gamma} P(y \mid x)$

   The estimator may be global / parametric or local / partition-based.

3. **Learn a decision score or boundary directly**

   Learn f(x); its sign or largest component determines the class.

   $f(x) \ \rightarrow\ h(x)$

   The learned score need not be a calibrated probability.

Examples of the Three Routes

Representative methods, grouped by the statistical quantity they aim to learn.

- **1. Class-conditional densities and priors**

  Naive Bayes, LDA, QDA

  Estimate p(x | y) and P(y), then apply Bayes' rule.

- **2. Posterior probability estimation**

  Global / parametric: logistic regression, softmax regression, softmax networks

  Local / partition: k-NN, classification trees, random forests

- **3. Direct decision-score learning**

  Perceptron, SVM, margin-trained neural networks

The category refers to the learning target; a model architecture may support more than one route.

Multi-class Classification

Generalize to the case that $\mathbf{y}$ take on more than two values

Theorem: Suppose that $\mathbf{y} \in \mathcal{Y} = \{1, \ldots, K\}$, the optimal rule is

$$h^*(x_0) = argmax_k\{P(\mathbf{y} = k|\mathbf{x} = x_0)\}$$

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Multi-class Classification

where

$$P(\mathbf{y} = k|\mathbf{x} = x_0) = \frac{f_k(x_0)\pi_k}{\sum_r f_r(x_0)\pi_r}$$

$$= \frac{P(\mathbf{x} = x_0|\mathbf{y} = 1)P(\mathbf{y} = 1)}{P(\mathbf{x} = x_0|\mathbf{y} = 1)P(\mathbf{y} = 1) + P(\mathbf{x} = x_0|\mathbf{y} = 0)P(\mathbf{y} = 0)}$$

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Toward LDA and QDA

The simplest approach to classification is to use the first approach and assume a parametric model for the densities.

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Linear Discriminant Analysis (LDA)

The simplest method is to use approach 1 and assume a parametric model for densities. Assume class conditional is Gaussian.

$\mathcal{Y} = \{0, 1\}$ assumed (i.e., 2 labels)

$$h(x) = \begin{cases} 1 & P(Y = 1|\mathbf{x} = x) > P(Y = 0|\mathbf{x} = x) \\ 0 & otherwise \end{cases}$$

$$P(Y = 1|\mathbf{x} = x) = \frac{f_1(x)\pi_1}{\sum_k f_k \pi_k} \quad \text{(denom = P(x))}$$

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Linear Discriminant Analysis (LDA)

1) Assume Gaussian distributions

$$f_k(x) = \frac{1}{(2\pi)^{d/2}|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_\mathbf{k})^T \Sigma_k^{-1} (\mathbf{x} - \mu_\mathbf{k})\right)$$

must compare

$\frac{f_1(x)\pi_1}{p(x)}$ with $\frac{f_0(x)\pi_0}{p(x)}$

Note that the p(x) denom can be ignored:

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Linear Discriminant Analysis (LDA)

$f_1(x)\pi_1$ with $f_0(x)\pi_0$

To find the decision boundary, set

$f_1(x)\pi_1 = f_0(x)\pi_0$

$$\frac{1}{(2\pi)^{d/2}|\Sigma_1|^{1/2}} exp(-\frac{1}{2}(\mathbf{x}-\mu_1)^T\Sigma_1^{-1}(\mathbf{x}-\mu_1))\pi_1 =$$
$$\frac{1}{(2\pi)^{d/2}|\Sigma_0|^{1/2}} exp(-\frac{1}{2}(\mathbf{x}-\mu_0)^T\Sigma_0^{-1}(\mathbf{x}-\mu_0))\pi_0$$

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Linear Discriminant Analysis (LDA)

2) Assume $\Sigma_1 = \Sigma_0$, we can use $\Sigma = \Sigma_0 = \Sigma_1$.

$$\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} exp(-\frac{1}{2}(\mathbf{x} - \mu_{\mathbf{1}})^T\Sigma^{-1}(\mathbf{x} - \mu_{\mathbf{1}}))\pi_1 =$$
$$\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} exp(-\frac{1}{2}(\mathbf{x} - \mu_{\mathbf{0}})^T\Sigma^{-1}(\mathbf{x} - \mu_{\mathbf{0}}))\pi_0$$

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Linear Discriminant Analysis (LDA)

3) Cancel $(2\pi)^{-d/2}|\Sigma|^{-1/2}$ from both sides.

$$exp(-\frac{1}{2}(\mathbf{x}-\mu_{\mathbf{1}})^T\Sigma^{-1}(\mathbf{x}-\mu_{\mathbf{1}}))\pi_1 = exp(-\frac{1}{2}(\mathbf{x}-\mu_{\mathbf{0}})^T\Sigma^{-1}(\mathbf{x}-\mu_{\mathbf{0}}))\pi_0$$

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Linear Discriminant Analysis (LDA)

4) Take log of both sides.

$$-\tfrac{1}{2}(\mathbf{x}-\mu_1)^T\Sigma^{-1}(\mathbf{x}-\mu_1)) + \log(\pi_1) =$$
$$-\tfrac{1}{2}(\mathbf{x}-\mu_0)^T\Sigma^{-1}(\mathbf{x}-\mu_0)) + \log(\pi_0)$$

5) Subtract one side from both sides, leaving zero on one side.

$$-\tfrac{1}{2}(\mathbf{x}-\mu_1)^T\Sigma^{-1}(\mathbf{x}-\mu_1) + \log(\pi_1) - [-\tfrac{1}{2}(\mathbf{x}-\mu_0)^T\Sigma^{-1}(\mathbf{x}-\mu_0) + \log(\pi_0)] = 0$$

$$\tfrac{1}{2}[-\mathbf{x}^T\Sigma^{-1}\mathbf{x} - \mu_1{}^T\Sigma^{-1}\mu_1 + 2\mu_1{}^T\Sigma^{-1}\mathbf{x} + \mathbf{x}^T\Sigma^{-1}\mathbf{x} + \mu_0{}^T\Sigma^{-1}\mu_0 - 2\mu_0{}^T\Sigma^{-1}\mathbf{x}] + \log(\tfrac{\pi_1}{\pi_0}) = 0$$

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Linear Discriminant Analysis (LDA)

Cancelling out the terms quadratic in $\mathbf{x}$ and rearranging results in

$$\frac{1}{2}[-\boldsymbol{\mu_1}^T\Sigma^{-1}\boldsymbol{\mu_1} + \boldsymbol{\mu_0}^T\Sigma^{-1}\boldsymbol{\mu_0} + (2\boldsymbol{\mu_1}^T\Sigma^{-1} - 2\boldsymbol{\mu_0}^T\Sigma^{-1})\mathbf{x}] + \log(\frac{\pi_1}{\pi_0}) = 0$$

We can see that the first pair of terms is constant, and the second pair is linear in x. Therefore, we end up with something of the form

$$\boldsymbol{a}^T\boldsymbol{x} + b = 0$$

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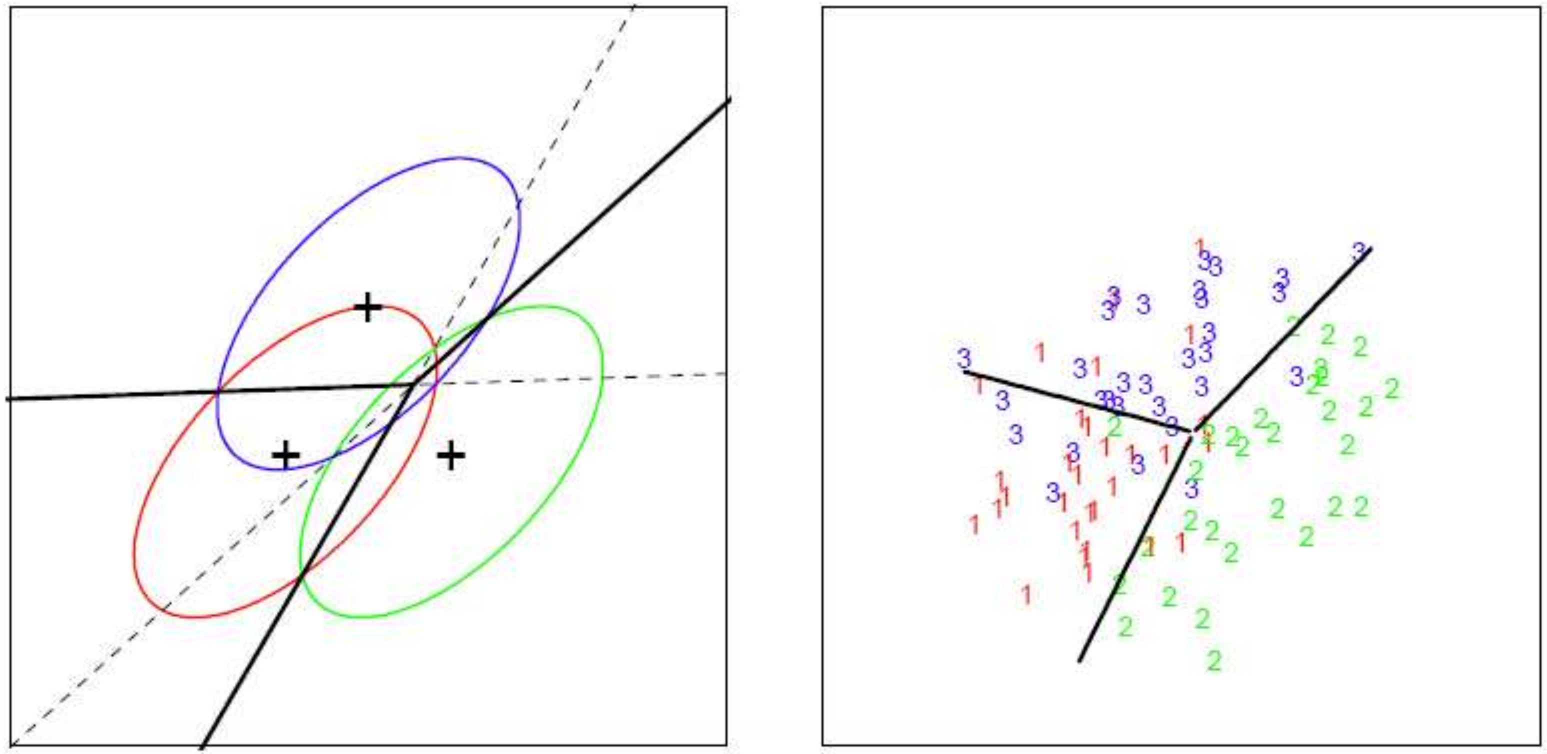

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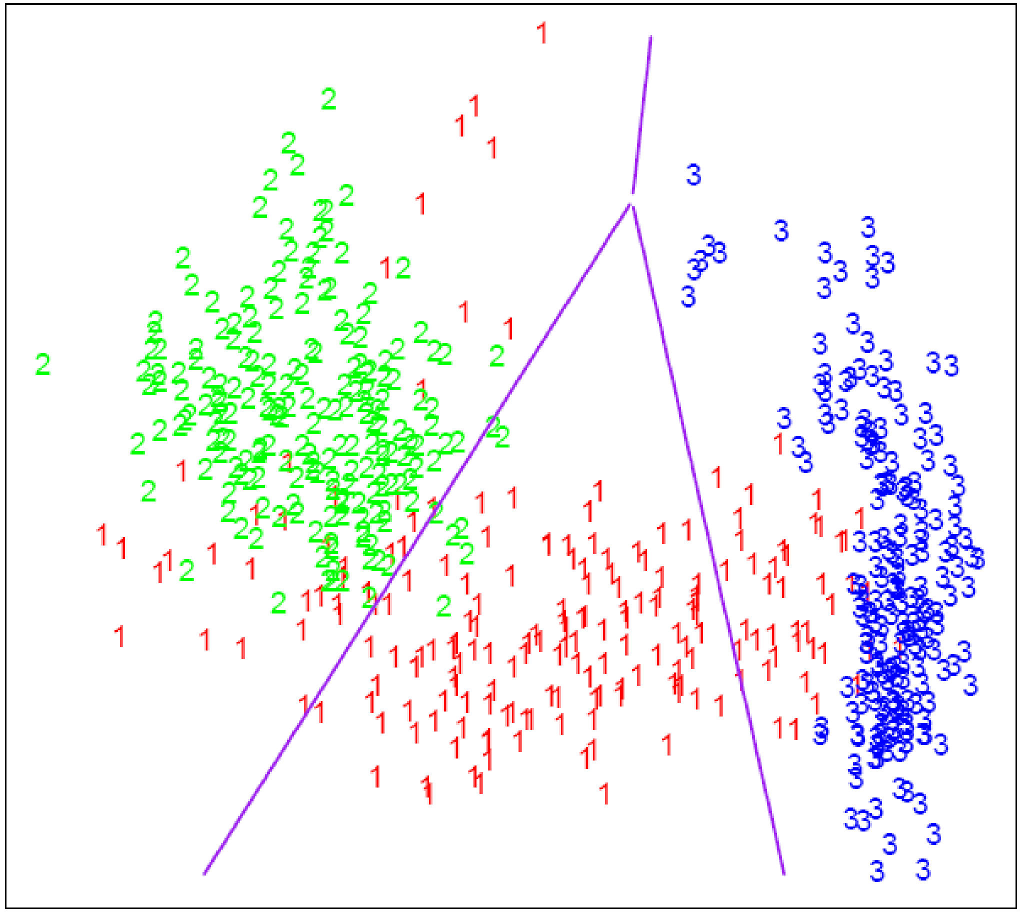
LDA



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Quadratic Discriminant Analysis (QDA)

If we relax assumption 2 (i.e. $\Sigma_1 \neq \Sigma_0$) then we get a quadratic equation that can be written as:

$$\boldsymbol{x}^T A \boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{x} + c = 0$$

Gaussian Class-Conditional Models Yield Quadratic Discriminant Functions

Suppose that $Y \in \{1, \ldots, K\}$, if $f_k(\mathbf{x}) = Pr(\mathbf{x} = \mathbf{x}|Y = k)$ is Gaussian. The Bayes Classifier is

$$h^*(\mathbf{x}) = \arg\max_k \delta_k(\mathbf{x})$$

Where

$$\delta_k(\mathbf{x}) = -\frac{1}{2}log(|\Sigma_k|) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) + log(\pi_k)$$

To compute this, we need to calculate the value for each class, and then take the one with the max value.

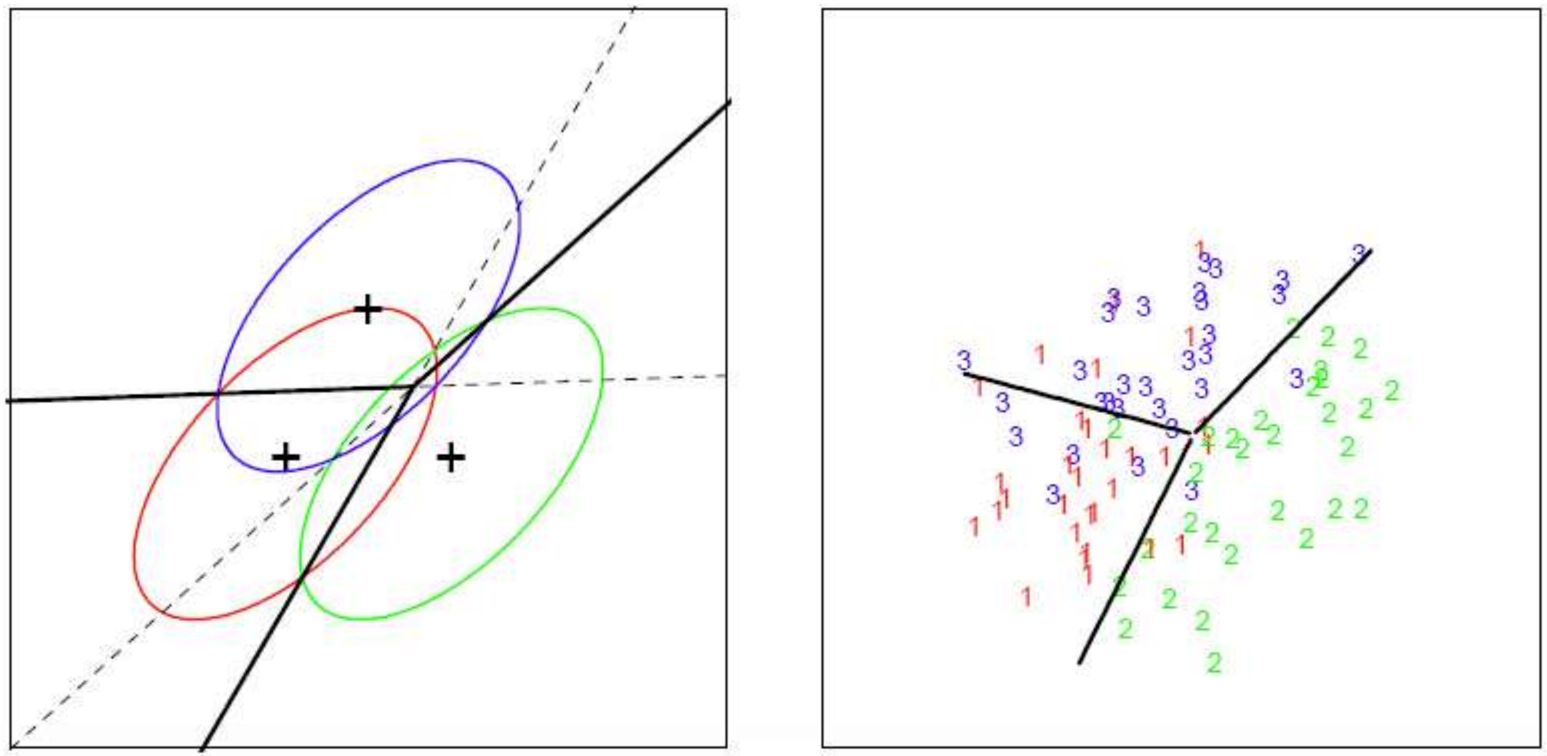
Shared Covariance Reduces QDA to LDA

When Gaussians have the same covariance

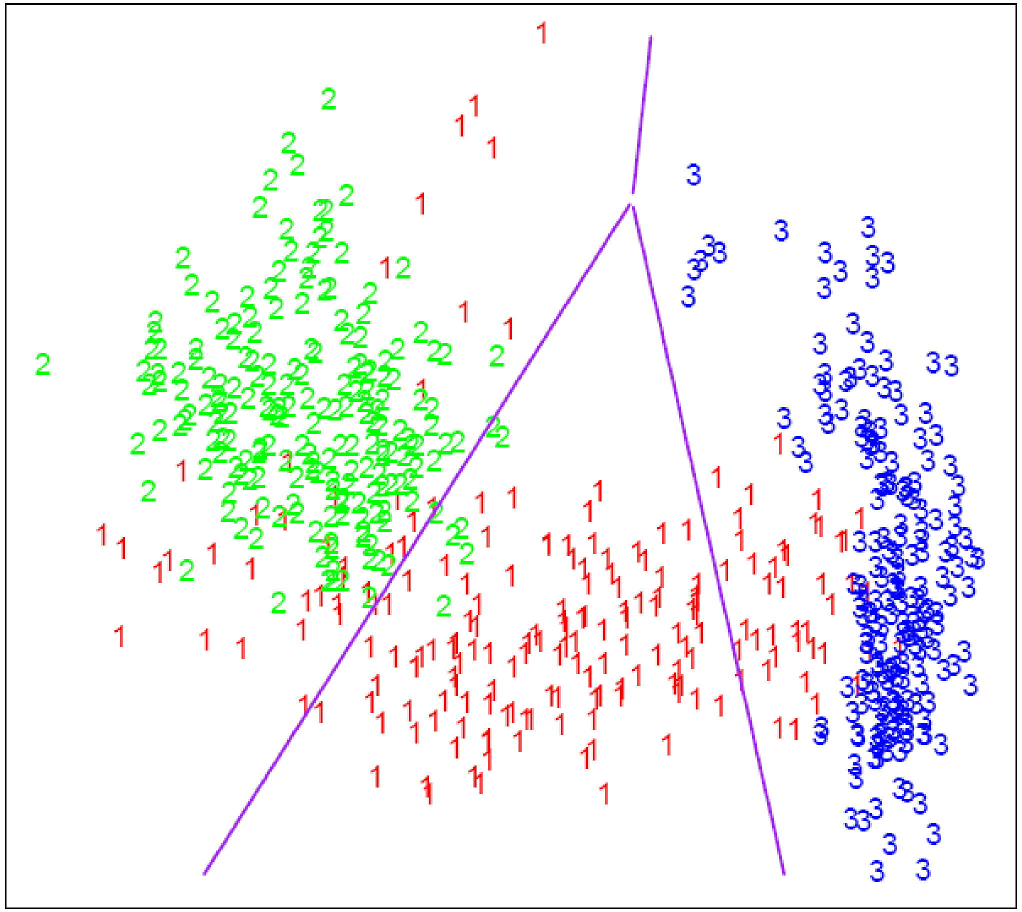
$$\Sigma_0 = \Sigma_1 = \Sigma_2 = \cdots = \Sigma_{k-1}$$

(e.g. LDA), then

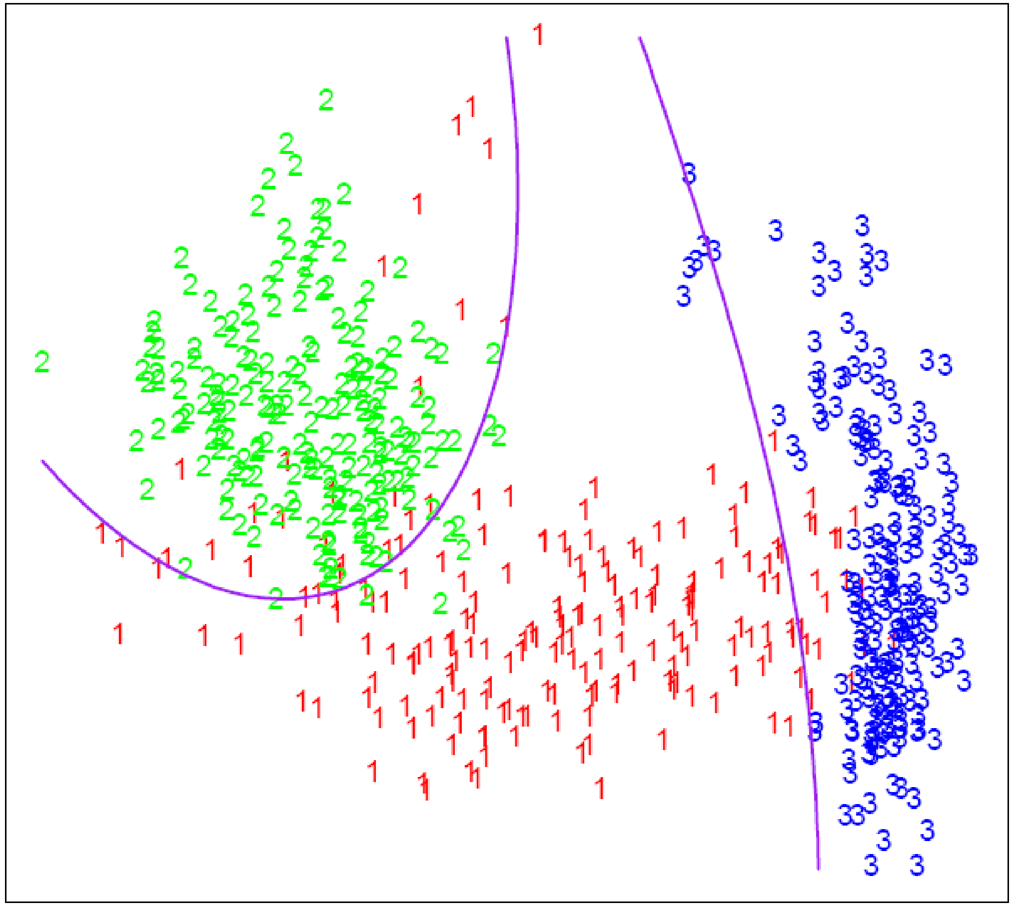
$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \tfrac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + log(\pi_k)$$

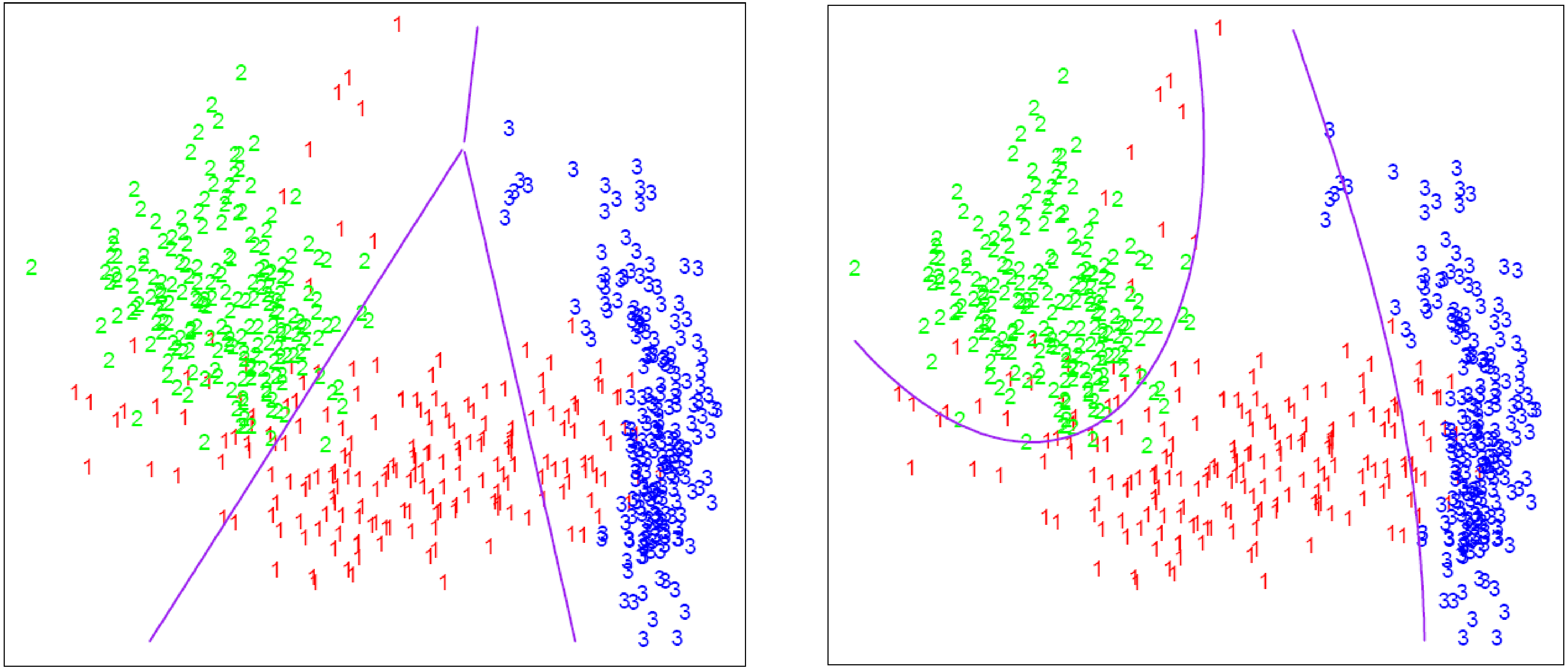


LDA



QDA





In practice

In practice we don't know the parameters of Gaussian and will need to estimate them using our training data.

$\hat{\pi}_k = \hat{P}r(y = k) = \frac{n_k}{n}$

where n_k is the number of class k observations.

$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$

$\hat{\Sigma}_k = \frac{1}{n_k - d} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^\top$

If we wish to use LDA we must calculate a common covariance, so we average all the covariances e.g.

$$\Sigma = \frac{\sum_{r=1}^{k}(n_r \Sigma_r)}{\sum_{r=1}^{k} n_r}$$

Where:

n_r is the number of data points in class r

Σ_r is the covariance of class r

n is the total number of data points, and k is the number of classes.

Alternative approach to do QDA

There is a trick that allows us to use the linear discriminant analysis (LDA) algorithm to generate a quadratic function that can be used to classify data. This trick is similar to, but more primitive than, the Kernel trick that will be discussed later in the course.

Alternative approach to do QDA

In this approach the feature vector is augmented with the quadratic terms (i.e. new dimensions are introduced) We then apply LDA on the new higher-dimensional data.

Alternative approach to do QDA

Using this trick, we introduce two new vectors, $\hat{\mathbf{w}}$ and $\hat{\mathbf{x}}$ such that:

$$\hat{\mathbf{w}} = [w_1, w_2, ..., w_d, v_1, v_2, ..., v_d]^T$$

and

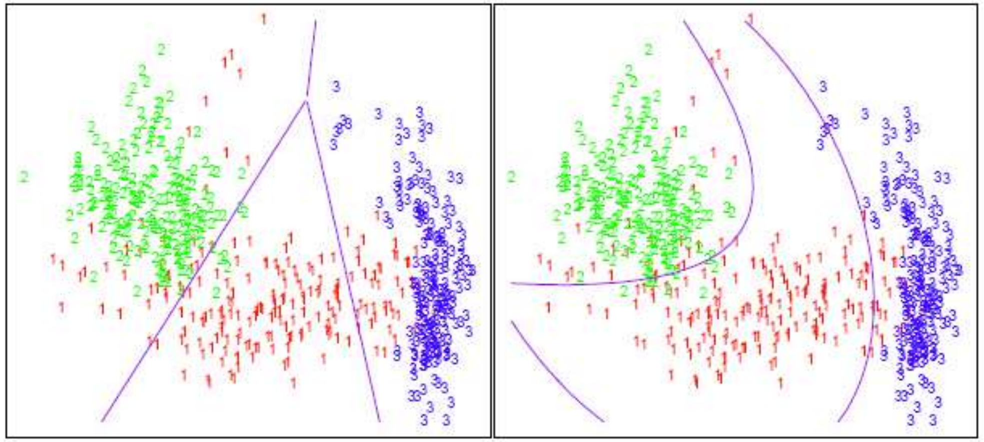
$$\hat{\mathbf{x}} = [x_1, x_2, ..., x_d, {x_1}^2, {x_2}^2, ..., {x_d}^2]^T$$

Alternative approach to do QDA

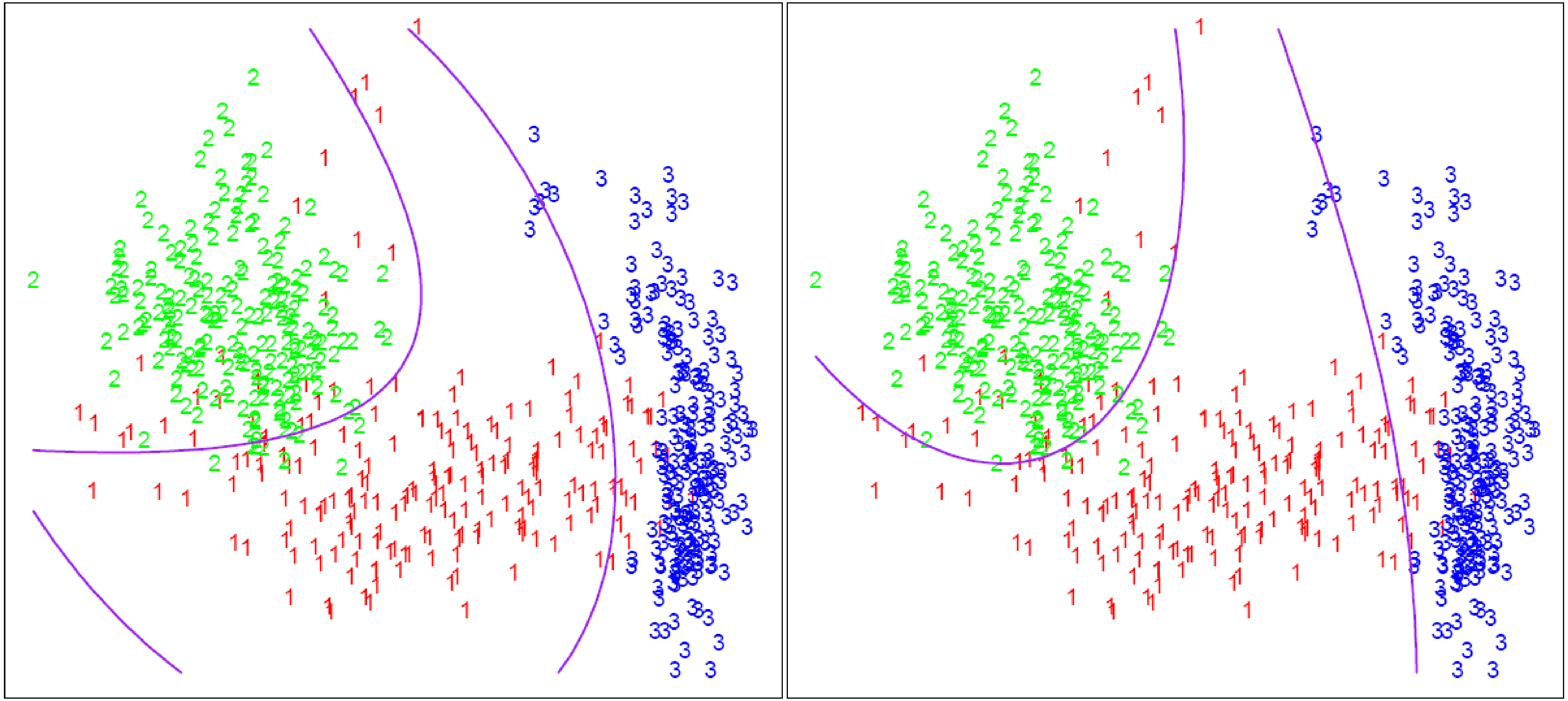
We can then apply LDA to estimate the new function:
$\hat{g}(\mathbf{x}, \mathbf{x}^2) = \hat{y} = \hat{\mathbf{w}}^T \hat{\mathbf{x}}$.

Note that we can do this for any x and in any dimension; we could extend a $d \times n$ matrix to a quadratic dimension by appending another $d \times n$ matrix with the original matrix squared, to a cubic dimension with the original matrix cubed, or even with a different function altogether, such as a $sin(x)$ dimension. Note, we are not applying QDA, but instead extending LDA to calculate a non-linear boundary, that will be different from QDA.

An alternative approach for QDA



The left plot shows some data from three classes, with linear decision boundaries found by linear discriminant analysis. The right plot shows quadratic decision boundaries. These were obtained by finding linear boundaries in the five-dimensional space $X_1, X_2, X_{12}, X_1^2, X_2^2$. *Linear inequalities in this space are quadratic inequalities in the original space.*



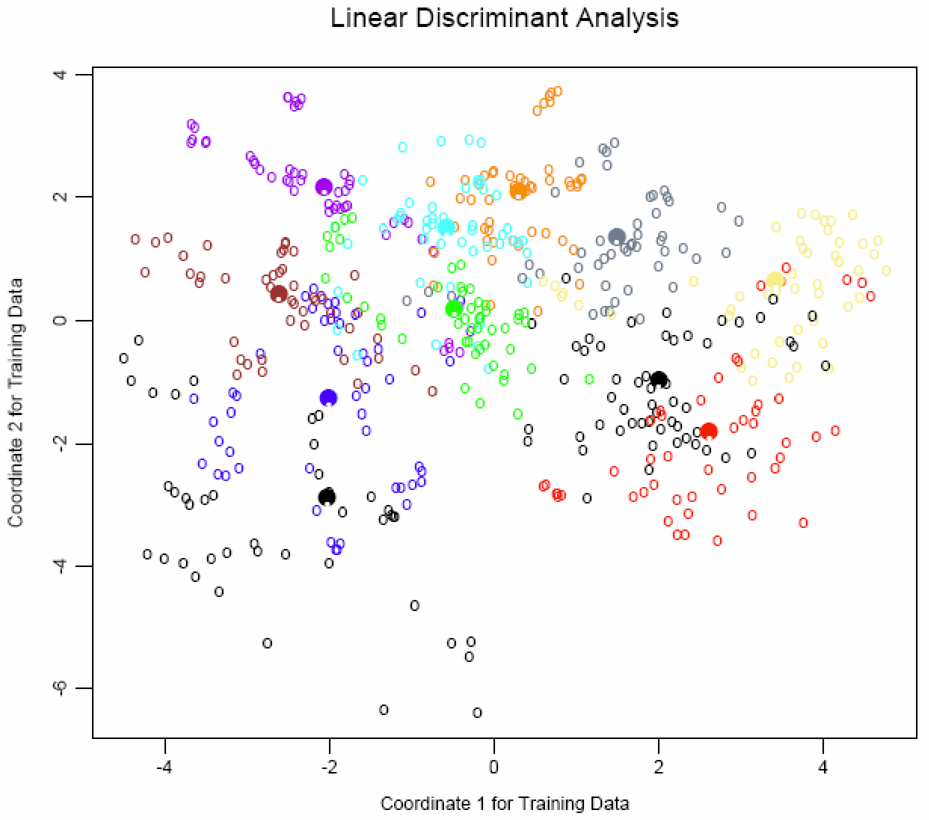
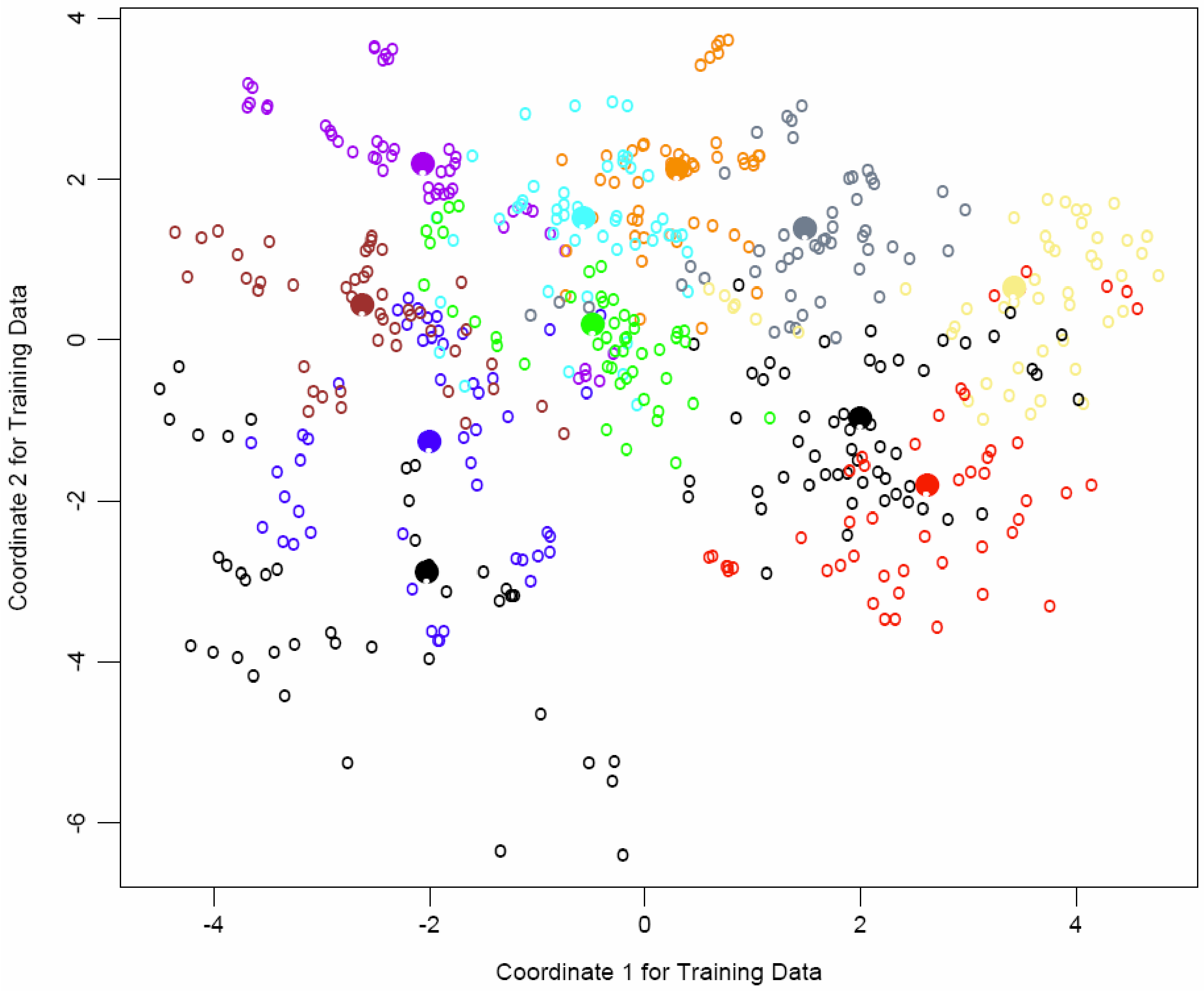
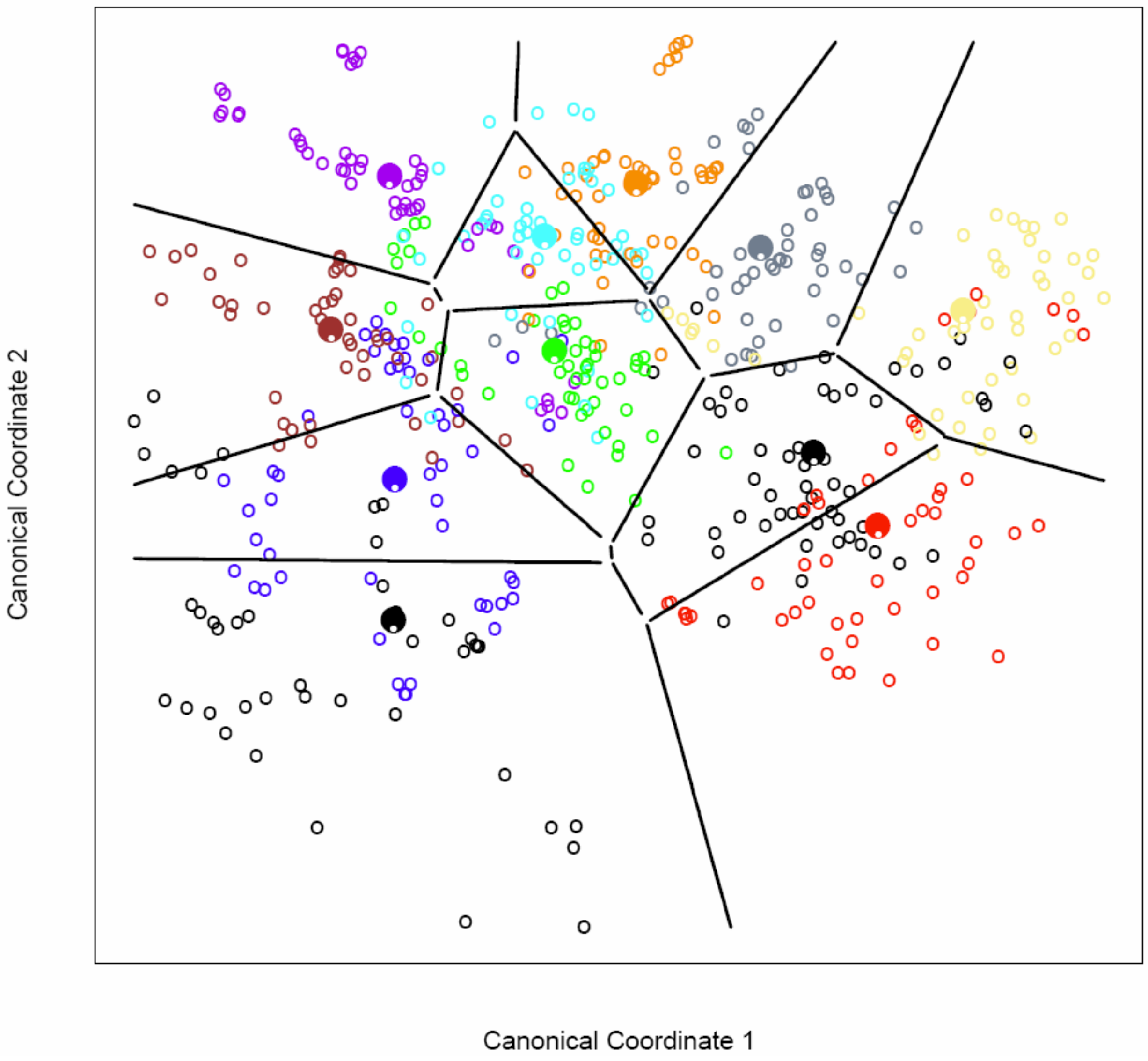


Figure 4.4: *A two-dimensional plot of the vowel training data. There are eleven classes with* $X \in \mathbb{R}^{10}$*, and this is the best view in terms of a LDA model (Section 4.3.3). The heavy circles are the projected mean vectors for each class. The class overlap is considerable.*

Linear Discriminant Analysis



Classification in Reduced Subspace



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Computation

For QDA we need to calculate:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}log(|\Sigma_k|) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) + log(\pi_k)$$

Lets first consider the case that

$$\Sigma_k = I, \forall k \; .$$

This is the case where each distribution is spherical, around the mean point.

Case 1

When $\Sigma_k = I$

We have:

$$\delta_k = -\frac{1}{2}log(|I|) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top I(\mathbf{x} - \boldsymbol{\mu}_k) + log(\pi_k)$$

but $\log(|I|) = \log(1) = 0$

and $(\mathbf{x} - \boldsymbol{\mu}_k)^\top I(\mathbf{x} - \boldsymbol{\mu}_k) = (\mathbf{x} - \boldsymbol{\mu}_k)^\top(\mathbf{x} - \boldsymbol{\mu}_k)$ is the squared Euclidean distance between two points $\mathbf{x}$ and $\boldsymbol{\mu}_k$

Thus under this condition (i.e. $\Sigma = I$) , a new point can be classified by its distance from the center of a class, adjusted by some prior.

Further, for two-class problem with equal prior, the discriminating function would be the perpendicular bisector of the 2-class's means.

Case 2

When $\Sigma_k \neq I$

Using the Singular Value Decomposition (SVD) of Σ_k

we get

$$\Sigma_k = U_k S_k U_k^\top$$

Note: Σ_k is a symmetric matrix $\Sigma_k = \Sigma_k^T$, so we have $\Sigma_k = U_k S_k U_k^\top$.

$$
\begin{aligned}
(\mathbf{x}-\boldsymbol{\mu}_k)^\top \Sigma_k^{-1}(\mathbf{x}-\boldsymbol{\mu}_k) \\
&= (\mathbf{x}-\boldsymbol{\mu}_k)^\top U_k S_k^{-1} U_k^T (\mathbf{x}-\boldsymbol{\mu}_k) \\
&= (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k)^\top S_k^{-1} (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k) \\
&= (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k)^\top S_k^{-\frac{1}{2}} S_k^{-\frac{1}{2}} (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k) \\
&= (S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k)^\top I (S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k) \\
&= (S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k)^\top (S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k)
\end{aligned}
$$

For δ_k , the second term becomes what is also known as the Mahalanobis distance:

Think of $S_k^{-\frac{1}{2}} U_k^\top$ as a linear transformation that takes points in class k and distributes them spherically around a point, like in Case 1.

Thus when we are given a new point, we can apply the modified δ_k values to calculate $h^*(x)$. After applying the singular value decomposition, Σ_k^{-1} is considered to be an identity matrix such that

$$\delta_k = -\frac{1}{2} log(|I|) - \frac{1}{2}[(S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k)^\top (S_k^{-\frac{1}{2}} U_k^\top \mathbf{x} - S_k^{-\frac{1}{2}} U_k^\top \boldsymbol{\mu}_k)] + log(\pi_k)$$

and,

$$\log(|I|) = \log(1) = 0$$

The difference between Case 1 and Case 2 (i.e. the difference between using the Euclidean and Mahalanobis distance) can be seen in the illu